Data 1030 Midterm Project Report

Liza Whitmire, Prasanna Nagarajan, & Declan Treacy
Brown University
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Introduction

The US Department of Agriculture (USDA) defines food insecurity as the status where households are uncertain about or unable to sufficiently meet the food needs of their family due to insufficient resources. According to the USDA, in 2024 86.3% of American households were food secure throughout the year and 13.7% (18.3 million) of households were food insecure at some point during the year (US Department of Agriculture, Economic Research Service, 2026b).

Our team chose to explore health factors predictive of food insecurity status among Americans. In the US, the rising cost of living has inhibited affordable access to food, with recent data revealing a 2.5% increase in food prices from September 2025 to September 2026 (USDA ERC, 2026a). While socioeconomic status (SES) is a well-documented predictor of food insecurity (Wight et al., 2016), there are a multitude of other predictors unrelated to economic wellbeing. Indeed, previous work has identified education, home ownership, age, unemployment, disability, state tax burden, and participation in nutrition assistance programs as predictors of food insecurity (Leitz, 2018). It is therefore important to emphasize that food insecurity is a multifaceted issue, with several economic, social, and systemic determinants, and negative impacts on physical and mental health and quality of life.

The consequences of food insecurity are well-documented. Among Americans, food insecurity is associated with increased risk of a multitude of adverse physical and mental health outcomes, including cardiovascular disease, diabetes, certain types of cancer, anxiety, depression, and chronic stress (Odoms-Young et al., 2023). Research on chronic food insecurity in children reveals that the consequences of food insecurity can persist throughout adulthood and inhibit children from reaching their full physical, cognitive, social, behavioral, and developmental potential, with consequences on academic performance, interpersonal skills (aggression, anxiety, friendships), and mental health (suicidal ideations, substance use disorders) (Gallegos et al., 2021; Shankar et al., 2017).

Food insecurity among Americans is therefore a compelling issue to investigate, particularly in the present sociopolitical environment where affordability and food access are at the forefront of national conversations. We are interested in leveraging data and modeling techniques to understand who is most vulnerable to food insecurity. Predicting food insecurity can create a better understanding of effective policy interventions, especially as food insecurity is a multidimensional issue with complex and interlinked determinants, as well as food crisis prediction which can improve response times for specific demographics in times of need. Recent work has demonstrated the efficacy of model-driven methods to predict the onset of food crisis using real-time market data, regional and geographic characteristics, and demographic information (Lentz et al., 2019), emphasizing the applicability of big data and machine learning techniques to predicting food security status.

Data

This dataset is sourced from the Integrated Public Use Microdata Series (Blewitt et al., 2026), which provides an interface for custom extracts of publicly available household- and individual-level census and survey data on social and economic features across global populations. Data is sourced from federal agencies, such as the Census Bureau and the Center for Disease Control (CDC). Our dataset is the National Health Interview Survey (NHIS), which is conducted annually by the CDC and collects information on health, healthcare access, and health-related behaviors among the civilian and non-institutionalized US population (Centers for Disease Control and Prevention, National Center for Health Statistics, 2024).

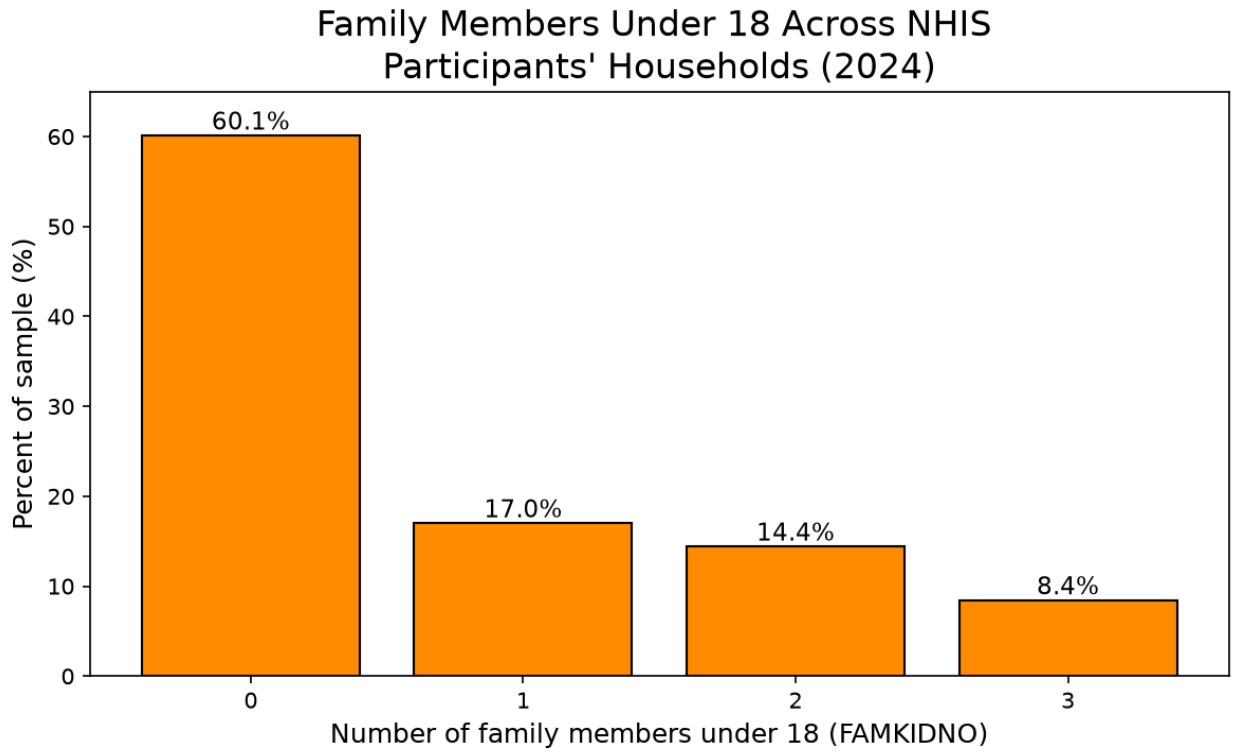
We are using the NHIS dataset from 2024, which surveyed 32,629 adults and 8,360 children for a total of 40,989 individual responses (rows). There are a total of 791 columns, including identification variables, which includes both individual- and household-level characteristics spanning several demographic, geographic, and health topics. Household variable topics include technical household information (e.g., survey year, unique ID) and household geography information. Individual variable topics include technical information, demographics, SES, general health, health conditions, COVID-19 conditions, health behaviors, health education, disability/limitations, medical care/access, health insurance, mental health, and vaccinations. Data types include continuous (e.g., BMI), categorical (e.g., ethnicity), and ordinal (e.g., educational attainment).

The target variable we chose is food security status for the household of the individual surveyed based on their raw score on the 30-day food security scale (FSSTATDET). The 30-day household food security scale (Nord, 2002) is a measure developed by the USDA Economic Research Service which assigns an integer value in the range 0 through 10 to the household based on the number of respondents' affirmative answers to questions on conditions, behaviors, and experiences within their household related to food insecurity within the previous 30 days. Lower scores indicate the family is food secure and higher scores indicate the family is food insecure. The food security status variable is constructed from responses to the 30-day household food security scale, dividing the scale into four categories: high food security (score of 0), marginal food security (a score of 1 or 2), low food security (a score of 3, 4, or 5), and very low food security (a score between 6 and 10). Within our dataset, 34,447 individuals are categorized as having high food security, 2,753 as having marginal food security, 2,165 as having low food security, and 1,565 as having very low food security. The food security statuses of 59 individuals are unknown.

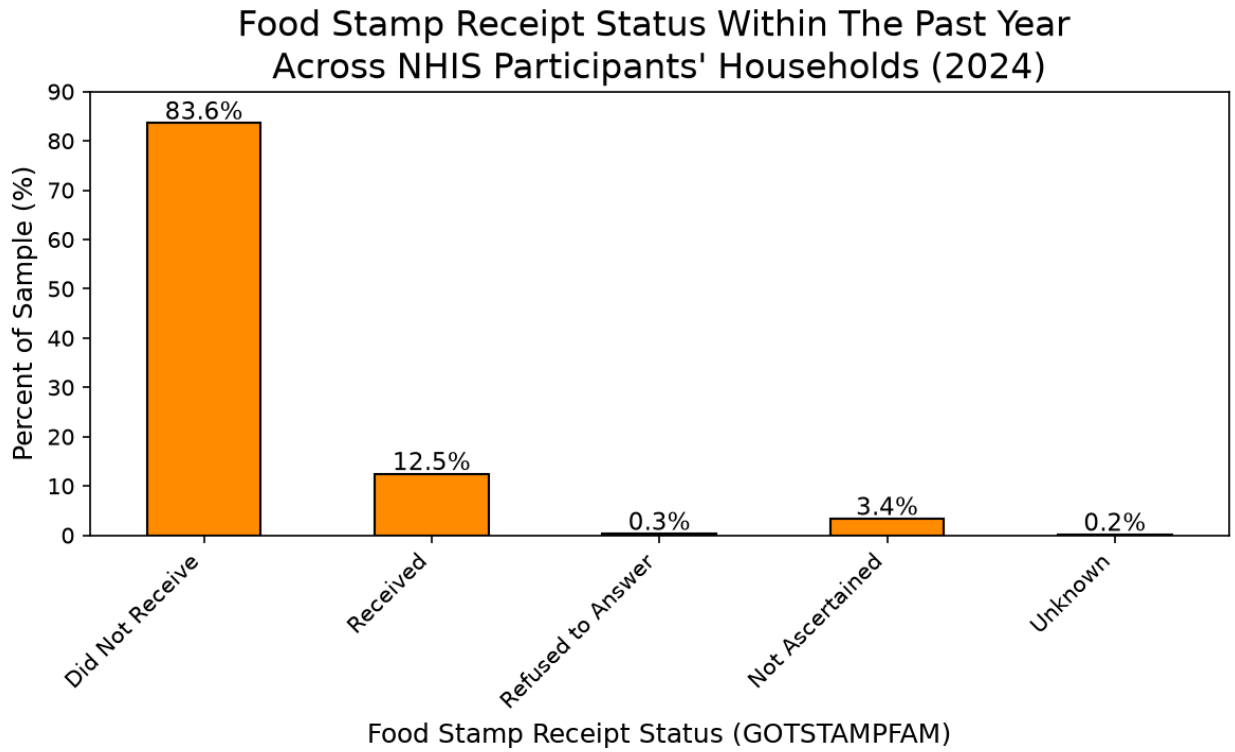
Exploratory Data Analysis

We conducted exploratory data analysis by checking variable type, number of unique values and their distribution, considering how to handle missing values, and its relationship to food security status. With previous work on predictors of food insecurity in mind, for our exploratory data figures, we choose to visualize the value counts of 3 variables that are related to or themselves have been identified as factors associated with food insecurity.

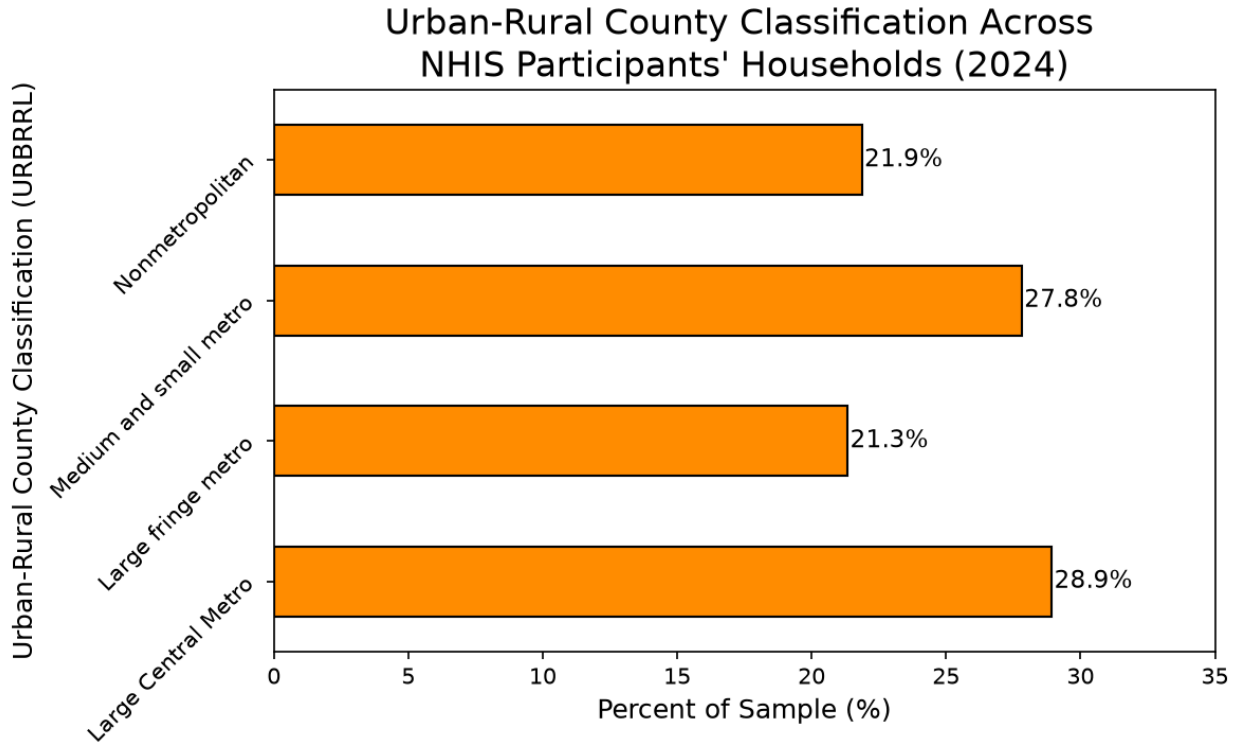
FAMKIDNO is a discrete variable describing how many family members under 18 live within the participant's household. We expect this variable to inform food security status, as more children within a family can mean a tighter food budget per person.



GOTSTAMPFAM is a categorical variable indicating whether anyone in the participant's family received food stamps or SNAP benefits during the previous calendar year. The Supplemental Nutrition Assistance Program (SNAP; previously the Food Stamps Program) is a federally-funded program which provides benefits to Americans in need of resources to afford nutritious food each month (US Department of Agriculture, Food and Nutrition Service). We expect this variable to inform food security status because participation in federal assistance programs likely indicates lack of sufficient resources to provide adequate food for an individual or family. Further, examining this variable may provide insight into how government assistance mitigates food insecurity.



URBRRL is an ordinal variable describing the urban–rural classification that categorizes the participant's household location into one of four groups: large central metro, large fringe metro, medium or small metro, and nonmetropolitan, based on the county where the household resides. We expect this variable to be informative because a family's geographic location impacts food availability and accessibility, two components of food security. For example, a household residing in a rural area may have a long drive to the nearest grocery store, limiting their access to varied, nutritious food.



Problem description

We aim to predict a household's food security status (very low, low, marginal, high) using demographic, socioeconomic, family, and health-related variables, thus comprising an ordinal classification problem. We were unable to find literature to-date using this specific dataset to predict food security status. Food insecurity is a widespread issue across our country with far-reaching consequences across an individual's physical and mental health and economic, educational, and social outcomes. The potential value of this project lies in improving understanding of which households are most vulnerable to food insecurity. By identifying the factors most strongly associated with our target variable, we can better understand drivers of food insecurity, in turn informing policies and programs aimed at supporting households most at risk of food insecurity. Potential users of this model may be healthcare professionals, community outreach workers, or government offices.

All features we will use in this project are based on survey responses that should be available at the time of deployment. We will drop variables that duplicate the same question in a different form, do not apply to all survey participants, or have a large share of missing or not applicable data.

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